Student Absenteeism and Academic Performance — Guiding Question 1

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Investigation Goal:

The goal of this investigation is to determine whether student absenteeism (categorized as fewer than 7 absences vs. more than 7 absences) is associated with student engagement and academic performance among students?

Guiding Q1)

Is the proportion of high-level performers in the low-absence group greater than the proportion of high-level performers in the high-absence group

```
# Loading the dataset

student_data = read.csv("dataset/xAPI-Edu-Data.csv")
dim(student_data)
```

```
## [1] 480  17
```

```
# Reading the dataset

head(student_data)
```

```
##   gender NationalITy PlaceofBirth    StageID GradeID SectionID Topic Semester
## 1      M          KW       KuwaIT lowerlevel    G-04         A    IT        F
## 2      M          KW       KuwaIT lowerlevel    G-04         A    IT        F
## 3      M          KW       KuwaIT lowerlevel    G-04         A    IT        F
## 4      M          KW       KuwaIT lowerlevel    G-04         A    IT        F
## 5      M          KW       KuwaIT lowerlevel    G-04         A    IT        F
## 6      F          KW       KuwaIT lowerlevel    G-04         A    IT        F
##   Relation raisedhands VisITedResources AnnouncementsView Discussion
## 1   Father          15               16                 2         20
## 2   Father          20               20                 3         25
## 3   Father          10                7                 0         30
## 4   Father          30               25                 5         35
## 5   Father          40               50                12         50
## 6   Father          42               30                13         70
##   ParentAnsweringSurvey ParentschoolSatisfaction StudentAbsenceDays Class
## 1                   Yes                     Good            Under-7     M
## 2                   Yes                     Good            Under-7     M
```

```
## 3                   No                      Bad            Above-7     L
## 4                   No                      Bad            Above-7     L
## 5                   No                      Bad            Above-7     M
## 6                  Yes                      Bad            Above-7     M
```

```
tail(student_data)
```

```
##     gender NationalITy PlaceofBirth      StageID GradeID SectionID     Topic
## 475      F      Jordan       Jordan MiddleSchool    G-08         A Chemistry
## 476      F      Jordan       Jordan MiddleSchool    G-08         A Chemistry
## 477      F      Jordan       Jordan MiddleSchool    G-08         A   Geology
## 478      F      Jordan       Jordan MiddleSchool    G-08         A   Geology
## 479      F      Jordan       Jordan MiddleSchool    G-08         A   History
## 480      F      Jordan       Jordan MiddleSchool    G-08         A   History
##     Semester Relation raisedhands VisITedResources AnnouncementsView Discussion
## 475        F   Father           2                7                 4          8
## 476        S   Father           5                4                 5          8
## 477        F   Father          50               77                14         28
## 478        S   Father          55               74                25         29
## 479        F   Father          30               17                14         57
## 480        S   Father          35               14                23         62
##     ParentAnsweringSurvey ParentschoolSatisfaction StudentAbsenceDays Class
## 475                    No                      Bad            Above-7     L
## 476                    No                      Bad            Above-7     L
## 477                    No                      Bad            Under-7     M
## 478                    No                      Bad            Under-7     M
## 479                    No                      Bad            Above-7     L
## 480                    No                      Bad            Above-7     L
```

```
table(student_data$StudentAbsenceDays, student_data$Class)
```

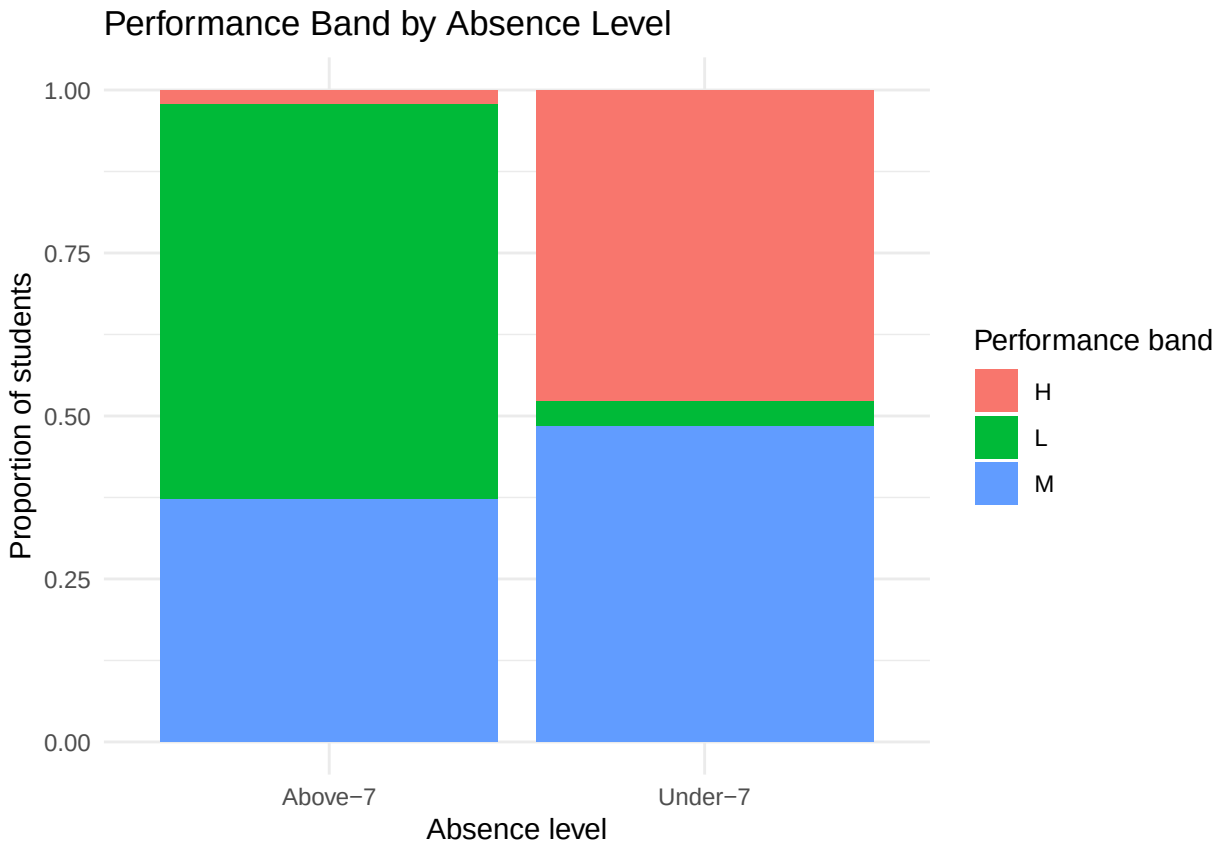
```
##
##             H   L   M
##   Above-7   4 116  71
##   Under-7 138  11 140
```

Interpretation

From the table above, we can observe that, out of 289 students in the low-absence group (Under-7), 138 have the High-performance band.

And, of the 191 students with high-absence group (Above-7), only 4 have reached the High-performance band

```
# Performance band by absence level
ggplot(student_data, aes(x = StudentAbsenceDays, fill = Class)) +
  geom_bar(position = "fill") +
  labs(x = "Absence level",
       y = "Proportion of students",
       fill = "Performance band",
       title = "Performance Band by Absence Level") +
  theme_minimal()
```




```
ggsave("figures/performance-by-absence.png", width = 7, height = 4.5, dpi = 150)
```

Stating the hypothesis:

Let p_{HN} be the proportion of students in the low-absence group (Under-7) who reach the High-performance band.

Let p_{HY} be the proportion of students in the high-absence group (Above-7) who reach the High-performance band.

$$H_0 : p_{\text{HN}} \leq p_{\text{HY}}$$

$$H_A : p_{\text{HN}} > p_{\text{HY}}$$

This indicates it is a right-tailed test.

We will be using a significance level of $\alpha = 0.05$

The Approach:

We are going to perform a permutation test on the difference in two sample proportions. The permutation test is condition-free and does not assume anything about the shape of the population distribution.

```r
# Creating a 1/0 data vector for High band performers

student_data = student_data %>% mutate(high_perf = ifelse(Class == "H", 1,0))

table(student_data$high_perf) # verifying the column and its values
```

```
##
##   0   1
## 338 142
```

```r
# Now, favstats orders groups alphabetically: [1] = Above-7, [2] = Under-7
# so mean[2] - mean[1] gives Under-7 minus Above-7, matching H_A
stats_gq1 = favstats(~high_perf | StudentAbsenceDays, data = student_data)
stats_gq1
```

```
##   StudentAbsenceDays min Q1 median Q3 max       mean        sd   n missing
## 1            Above-7   0  0      0  0   1 0.02094241 0.1435679 191       0
## 2            Under-7   0  0      0  1   1 0.47750865 0.5003603 289       0
```

```r
obs_diff = stats_gq1$mean[2] - stats_gq1$mean[1]
print(obs_diff)
```

```
## [1] 0.4565662
```

```r
# Now, permute


n_total = nrow(student_data)
n_under = 289
N = 1999
perm.outcome = numeric(N)

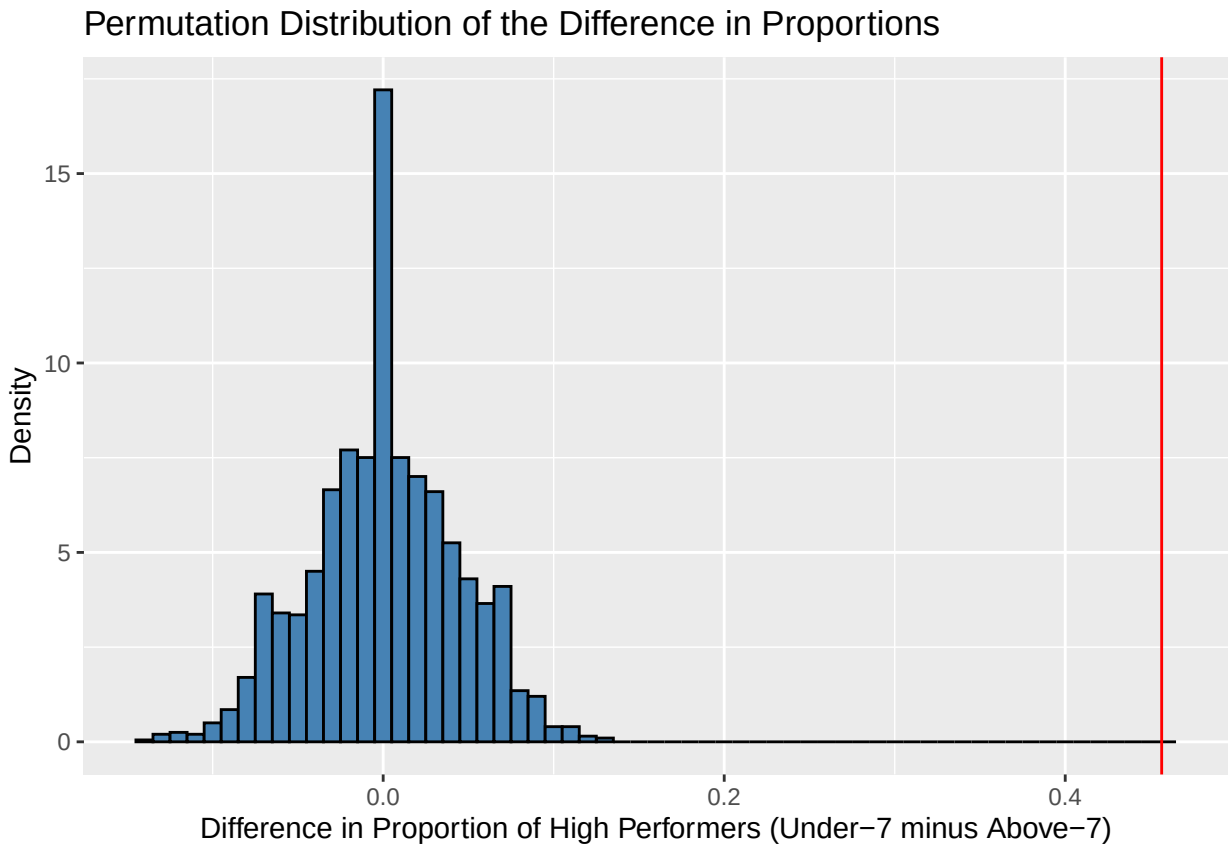
for(i in 1:N)
{
  index = sample(n_total, n_under, replace = FALSE)
  perm.outcome[i] = mean(student_data$high_perf[index]) - mean(student_data$high_perf[-index])
}

perm.df = data.frame(perm.outcome)
summary(perm.df)
```

```
##   perm.outcome
##  Min.   :-0.1434446
##  1st Qu.:-0.0303991
##  Median : 0.0043841
##  Mean   : 0.0006735
##  3rd Qu.: 0.0304716
##  Max.   : 0.1348213
```

```r
# Now, visualizing the permutation distribution using histogram

ggplot(perm.df, aes(x = perm.outcome)) + geom_histogram(aes(y = after_stat(density)), col = 'black', fi
  ggtitle("Permutation Distribution of the Difference in Proportions")
```




```r
ggsave("figures/permutation-distribution.png", width = 8, height = 5, dpi = 150)
```

Interpretation

What we can infer from this histogram plot is: The histogram shows the distribution of the difference in proportions produced by 1999 random permutations of the absence labels. It is centred near zero and ranges from about -0.10 to +0.15, which is what we would expect if absenteeism had no bearing on performance. The red line indicates that the observed difference of 0.4566, which lies far outside the range of anything the permutations produced.

```r
# Now, computing the P-value

emp_pvalue_gq1 = (sum(perm.outcome >= obs_diff) + 1) / (N + 1)
emp_pvalue_gq1
```

```
## [1] 5e-04
```

```r
# Now finally, computing the 95% confidence-interval for the difference in the proportions

x_HN = 138
x_HY = 4

n_HN = 289
n_HY = 191

prop.test(c(x_HN +1, x_HY + 1) , c(n_HN + 2, n_HY + 2), alternative = "two.sided", correct = FALSE)$con
```

```
## [1] 0.3901454 0.5133676
## attr(,"conf.level")
## [1] 0.95
```

Interpretation

What we can infer from the results obtained is:

Observed difference = : $\hat{p}_{\text{HN}} - \hat{p}_{\text{HY}} = 0.4566$ Empirical P-value: 0.0005

This means that if absenteeism had no bearing on reaching the High-performance band, so that the two group labels carried no meaning, then the probability of getting a difference in proportions as large as the observed 0.4566 is about 0.0005. Which is extremely small, and a difference this large is very unlikely to happen by chance alone.

Decision: Since the empirical p-value is 0.0005, which is less than $\alpha = 0.05$, we reject H_0.

Conclusion: From these results, there is very strong statistical evidence that the proportion of students reaching the High-performance band is greater among students with fewer than 7 absence days than among students with more than 7 absence days.

95% Confidence Interval:

Also, we estimate with 95% confidence that the proportion of students reaching the High-performance band is between 0.3901 and 0.5134 higher in the low-absence group than in the high-absence group. Since the interval lies entirely above zero, it is consistent with the decision to reject H_0.

$$0.3901 < p_{\text{HN}} - p_{\text{HY}} < 0.5134$$
